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Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$\begin{aligned} h = \tan(\varphi) &= \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} &= \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \end{aligned} \quad (1)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (2)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (3)$$

V_0 = Opposing Potential ($I = 0$)

$$\varphi = \angle(E_{kin}, \nu) \quad e = \text{electron charge} \quad Q = \text{ion charge}$$

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

¹Wirkungsquantum